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摘　要

Conventional product development employs a design-build-test philosophy. The sequentially executed development process often results in prolonged lead times and elevated product costs. The proposed e-Design paradigm employs IT-enabled technology for product design, including virtual prototyping (VP) to support a cross-functional team in analyzing product performance, reliability, and manufacturing costs early in product development, and in making quantitative trade-offs for design decision making. Physical prototypes of the product design are then produced using the rapid prototyping (RP) technique and computer numerical control (CNC) to support design verification and functional prototyping, respectively.

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Abstract

Conventional product development employs a design-build-test philosophy. The sequentially executed development process often results in prolonged lead times and elevated product costs. The proposed e-Design paradigm employs IT-enabled technology for product design, including virtual prototyping (VP) to support a cross-functional team in analyzing product performance, reliability, and manufacturing costs early in product development, and in making quantitative trade-offs for design decision making. Physical prototypes of the product design are then produced using the rapid prototyping (RP) technique and computer numerical control (CNC) to support design verification and functional prototyping, respectively.

Key Words: Computer-Aided Design; FEM; CAM

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Chapter 1 Introduction

Conventional product development is a design-build-test process. Product performance and reliability assessments depend heavily on physical tests, which involve fabricating functional prototypes of the product and usually lengthy and expensive physical tests. Fabricating prototypes usually involves manufacturing process planning and fixtures and tooling for a very small amount of production. The process can be expensive and lengthy, especially when a design change is requested to correct problems found in physical tests.

1.1 Introduction

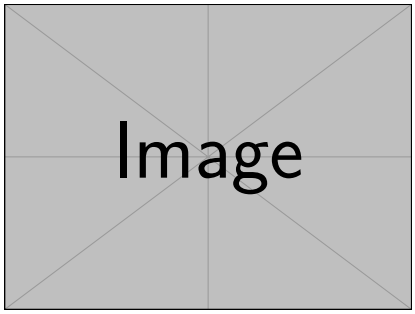

Figure 1-1 Comparison of the percentages of customer needs that are revealed for focus groups and interviews as a function of the number of sessions

In conventional product development, design and manufacturing tend to be disjointed. Often, manufacturability of a product is not considered in design. Manufacturing issues usually appear when the design is finalized and tests are completed. Design defects related to manufacturing in process planning or production are usually found too late to be corrected. Consequently, more manufacturing procedures are necessary for production, resulting in elevated product cost[1]. With this highly structured and sequential process, the product development cycle tends to be extended, cost is elevated, and product quality is often compromised to avoid further delay. Costs and the number of engineering change requests (ECRs) throughout the product development cycle are often proportional according to the pattern shown in Figure 1-1.

It is reported that only 8% of the total product budget is spent for design; however,

in the early stage, design determines 80% of the lifetime cost of the product (Anderson 1990). Realistically, today's industries will not survive worldwide competition unless they introduce new products of better quality, at lower cost, and with shorter lead times. Many approaches and concepts have been proposed over the years, all with a common goal to shorten the product development cycle, improve product quality, and reduce product cost[2].

1.2 Background

A number of proposed approaches are along the lines of virtual prototyping, which is a simulation-based method that helps engineers understand product behavior and make design decisions in a virtual environment. The virtual environment is a computational framework in which the geometric and physical properties of products are accurately simulated and represented. A number of successful virtual prototypes have been reported, such as Boeing's 777 jetliner, General Motors' locomotive engine, Chrysler's automotive interior design, and the Stockholm Metro's Car 2000. In addition to virtual prototyping, the concurrent engineering (CE) concept and methodology have been studied and developed with emphasis on subjects such as product life cycle design, design for X-abilities (DFX), integrated product and process development (IPPD), and Six Sigma.

Chapter 2 Identifying Customer Needs

A successful hand tool manufacturer was exploring the growing market for handheld power tools. After performing initial research, the firm decided to enter the market with a cordless screwdriver. Exhibit 5-1 shows several existing products used to drive screws. After some initial concept work, the manufacturer's development team fabricated and field-tested several prototypes. The results were discouraging. Although some of the products were liked better than others, each one had some feature that customers objected to in one way or another. The results were quite mystifying since the company had been successful in related consumer products for years. After much discussion, the team decided that its process for identifying customer needs was inadequate.

2.1 Method of study

This chapter presents a method for comprehensively identifying a set of customer needs. The goals of the method are to:

- Ensure that the product is focused on customer needs.
- Identify latent or hidden needs as well as explicit needs.
- Provide a fact base for justifying the product specifications.
- Create an archival record of the needs activity of the development process.
- Ensure that no critical customer need is missed or forgotten.
- Develop a common understanding of customer needs among members of the development team.

The philosophy behind the method is to create a high-quality information channel that runs directly between customers in the target market and the developers of the product. This philosophy is built on the premise that those who directly control the details of the product, including the engineers and industrial designers, must interact with customers and experience the use environment of the product. Without this direct experience, technical trade-offs

are not likely to be made correctly, innovative solutions to customer needs may never be discovered, and the development team may never develop a deep commitment to meeting customer needs. See table 2-1

Table 2-1 Range of variables in design of experiment

	AAA	**BBB**	**CCC**	**DDD**
Alpha	1000	10000	500	50%
Beta	5500	5000	220	22%
Gamma	1100	1000	280	28%
Sum	17600	16000	1000	100%

2.1.1 Process of identifying customer needs

The process of identifying customer needs is an integral part of the larger product development process and is most closely related to concept generation, concept selection, competitive benchmarking, and the establishment of product specifications. The customer-needs activity is shown in Exhibit 5-2 in relation to these other front-end product development activities, which collectively can be thought of as the concept development phase.

2.1.2 Development process

The concept development process illustrated in Exhibit 5-2 implies a distinction between customer needs and product specifications. This distinction is subtle but important.

Chapter 3 Engineering Design

Although engineering drawing still plays an important role in product design and manufacturing in many industrial sectors around the world, manual sketching for creating drawings has been gradually replaced by CAD (computer-aided design) software using computers. Beginning in the 1980s, CAD software reduced the need for draftsmen significantly, especially in small to mid-sized companies. The software's affordability and ability to run on personal computers in the mid-1990s allowed engineers to do their own drafting and analytic work to some extent 3-1.

$$x^n + y^n = z^n \tag{3-1}$$

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```
``No, sir!'' All the pedant in Dean Hart was aroused.
``In the English language, the word `majority' means `more than half'.
Thirteen out of fourteen is a majority, nothing more.''

“他不回答，对柜里说，‘温两碗酒，要一碟茴香豆。’便排出九文大钱。”
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Conclusions

During the 1980s, design-for-manufacturing practices were put into place in thousands of firms. Today DFM is an essential part of almost every product development effort. No longer can designers "throw the design over the wall" to production engineers. As a result of this emphasis on improved design quality, some manufacturers claim to have reduced production costs of products by up to 50 percent. In fact, comparing current new product designs with earlier generations, one can usually identify fewer parts in the new product, as well as new materials, more integrated and custom parts, higher-volume standard parts and subassemblies, and simpler assembly procedures.

References

[1] 袁庆龙, 章军, 侯文义. 热处理温度对 ni-p 合金刷镀层组织形貌及显微硬度的影响[J]. 太原理工大学学报, 2001(01): 54-56.

[2] 罗云. 安全科学理论体系的发展及趋势探讨[J]. 兵工安全技术, 1999(03): 22-25.

Appendices

Appendix A Design Structure Matrix Example

One of the most useful applications of the design structure matrix (DSM) method is to represent well-established, but complex, engineering design processes. This rich process modeling approach facilitates:

- Understanding of the existing development process.
- Communication of the process to the people involved.
- Process improvement.
- Visualization of progress during the project.

Exhibit 18-14 shows a DSM model of a critical portion of the development process at a major automobile manufacturer.

Appendix B Image Test

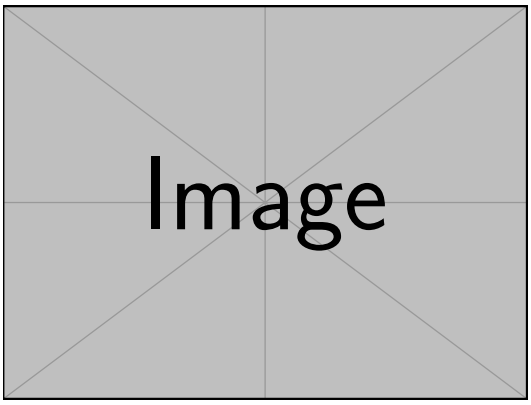

Figure B-1 Hello World

Some text that ref to the figure B-1.

Acknowledgements

This book contains material developed for use in the interdisciplinary courses on product development that we teach. Participants in these courses include graduate students in engineering, industrial design students, and MBA students. While we aimed the book at interdisciplinary graduate-level audiences such as this, many faculty teaching graduate and undergraduate courses in engineering design have also found the material useful. Product Design and Development is also for practicing professionals. Indeed, we could not avoid writing for a professional audience, because most of our students are themselves professionals who have worked either in product development or in closely related functions.

This book blends the perspectives of marketing, design, and manufacturing into a single approach to product development. As a result, we provide students of all kinds with an appreciation for the realities of industrial practice and for the complex and essential roles played by the various members of product development teams. For industrial practitioners, in particular, we provide a set of product development methods that can be put into immediate practice on development projects.